

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Applied ML Aug 2025 to May 2026

- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.
- Ran **evals** on RAG workflows by comparing chunking strategies, embedding models, and top-k settings, and classified each failure as retrieval-caused or generation-caused.
- Boosted malware classification to **80.2%** accuracy across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.

Turito | Software Engineering Intern, Backend Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

LLM & AI: RAG, embeddings, evals, agents, tool use, LangChain, OpenAI and Anthropic APIs, PyTorch, Hugging Face

Backend & Frontend: FastAPI, asyncio, SSE, REST, gRPC, React, Next.js, Node.js

Data & Infra: PostgreSQL, Redis, Qdrant, SQLite, Kafka, Temporal, Docker, AWS, pytest

PROJECTS

Document Action Agent | *React, TypeScript, FastAPI, LLMs*

- Built a document-grounded assistant that extracts key details, verifies sources before recommending actions, and routes ambiguous cases to **human review**.

Kairo, Patient Assistant | *React, TypeScript, FastAPI, OpenAI, Vapi, AWS*

- Shipped a chat and voice patient-service assistant with LLM routing and voice-call handoff in **under one week**.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.

OPEN SOURCE

- **corsairdev/corsair**: 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed
- **microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Distributed Computing, Database Systems, Advanced Algorithms, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)